SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.Tech-AIDS/2024	**Date:**	

Code of Conduct

I S.Krishna Chaithanya(192424287)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Krishna Chaithanya

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
   A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.
   An Industry 4.0 manufacturing plant monitors machines using semantic rules.
   Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $Active(x) \rightarrow Producing(x)$
- $Produces(x,y) \wedge Active(x) \rightarrow Available(y)$
- $Maintenance(x) \rightarrow \neg Producing(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines.
   An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand

"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	**Clicked Result**
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	**Parse Structure**
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	**Role**
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Q1. Semantic Representation in Customer Support Chatbots

1. Action–Object Relationships in the Semantic Representations

Each customer query is converted by the semantic analyzer into a predicate-argument structure of the general form ACTION(Object/Service, Agent). The predicate names the intent (action) the customer wants performed, the first argument identifies the entity/service being acted upon, and the second argument identifies the agent (Customer) initiating the request.

Customer Query	Semantic Representation	Action (Predicate)	Object	Agent
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)	ACTIVATE	Roaming	Customer
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)	DEACTIVATE	CallerTune	Customer
"Check my data balance."	QUERY(DataBalance, Customer)	QUERY	DataBalance	Customer
"Enable 5G service."	ACTIVATE(5GService, Customer)	ACTIVATE	5GService	Customer

Structurally, all four representations share the same two-place predicate schema, which shows that the semantic analyzer correctly captures the argument structure (who does what to which service) even when it fails to capture the correct action itself (see Section 2). This distinction between correct structure and correct content is important**: a chatbot can be syntactically/structurally consistent while still being semantically wrong.**

2. Query Containing Semantic Interpretation Errors

Query ID	Actual User Intent	Predicted Intent	Match?
Q1	Activate Roaming	Activate Roaming	Correct
Q2	Deactivate Caller Tune	Activate Caller Tune	INCORRECT
Q3	Check Data Balance	Query Data Balance	Correct (paraphrase)
Q4	Enable 5G Service	Activate 5G Service	Correct (synonym)

Query Q2 contains a clear semantic interpretation error. The actual intent is DEACTIVATE(CallerTune, Customer), but the system predicted ACTIVATE(CallerTune, Customer) — the polarity of the action predicate has been reversed. This is not a minor lexical mismatch (as in Q3/Q4, where "Check" ≈ "Query" and "Enable" ≈ "Activate" are synonymous and semantically equivalent); it is an antonym substitution error, where the analyzer confused two opposite predicates. This type of error is the most dangerous class of semantic error because it produces a well-formed, confident output that is the exact opposite of what the user wants.

3.Impact of Meaning Representation on Chatbot Decision Accuracy

- Action Reversal Risk: As seen in Q2, a single predicate-polarity error (ACTIVATE vs DEACTIVATE) causes the chatbot to perform the opposite of the customer's request, leading to service disruption, billing complaints, and loss of customer trust.
- Cascading Downstream Failures: Since the semantic representation is used directly to trigger backend APIs (e.g., activateService(), deactivateService()), an interpretation error is not just a conversational misunderstanding — it becomes an incorrect real-world system action.
- Synonym Robustness: Q3 and Q4 show that the analyzer is robust to lexical variation (QUERY vs Check, ACTIVATE vs Enable), which is desirable and indicates a reasonably good semantic normalization layer.
- Precision vs. Recall Trade-off: A chatbot tuned for high recall (always classifying an intent) without corresponding attention to precision on antonymic action pairs will systematically misfire on toggle-type commands (activate/deactivate, enable/disable, subscribe/unsubscribe).
- Customer Escalation Cost: Misclassified toggle actions typically require human agent intervention **to reverse, increasing operational cost and reducing the chatbot's automation rate.**

4.Recommendations to Improve Semantic Understanding

- Contrastive Predicate Training: Explicitly train the intent classifier on minimal pairs (activate/deactivate, enable/disable, subscribe/unsubscribe) so the model learns to weight negation/polarity cue words ("deactivate", "stop", "cancel", "turn off") strongly.

- Negation and Polarity Detection Module: Add a dedicated polarity-detection sub-component that runs before intent classification and flags the presence of negating or reversing verbs, then cross-checks the final predicate against the detected polarity.
- Confidence Thresholding with Confirmation: For any toggle-type action predicate, require an explicit confirmation step ("You want to deactivate Caller Tune — confirm?") before executing the backend action, especially when model confidence is below a defined threshold.
- Semantic Role Labelling (SRL) Integration: Combine intent classification with SRL to independently verify the Agent, Action, and Theme roles, catching cases where the action slot is filled incorrectly despite correct entity extraction.
- Continuous Feedback Loop: Log all instances where the predicted action differs from the action ultimately confirmed/corrected by the user, and use this data to retrain and re-calibrate the semantic analyzer periodically.
- Ontology-Backed Validation: Maintain a service ontology that defines valid state transitions (e.g., a service can only be deactivated if it is currently active) and use it to sanity-check predicted actions against current account state.

Q2. First-Order Predicate Calculus for Smart Manufacturing

1. FOPC Representation of the Production Data
Ground facts derived from the production data table:

- Active(M1)
- Active(M2)
- Maintenance(M3)
- Active(M4)

Given predicate rules (universally quantified over machines x and products y):

- $\forall x\ \ Active(x) \rightarrow Producing(x)$
- $\forall x \forall y\ \ Produces(x,y) \land Active(x) \rightarrow Available(y)$
- $\forall x\ \ Maintenance(x) \rightarrow \neg Producing(x)$

To connect machines to the products they manufacture (needed for Task 2 below), we introduce the standard shop-floor mapping implied by the scenario — each machine produces one product line:

- Produces(M1, ComponentA)
- Produces(M2, ComponentB)
- Produces(M3, Gear)
- Produces(M4, ComponentC)

2. Inference — Which Products Are Currently Available

Applying Rule 1 ($Active(x) \rightarrow Producing(x)$) to the ground facts:

- $Active(M1) \rightarrow Producing(M1)$
- $Active(M2) \rightarrow Producing(M2)$
- $Active(M4) \rightarrow Producing(M4)$
- $Maintenance(M3) \rightarrow \neg Producing(M3)$ (via Rule 3)

Applying Rule 2 ($Produces(x,y) \land Active(x) \rightarrow Available(y)$) using the Produces facts:

Machine	Status	Producing?	Product	Available(y)?
M1	Active	Yes	ComponentA	Available(ComponentA) — TRUE
M2	Active	Yes	ComponentB	Available(ComponentB) — TRUE
M3	Maintenance	No (¬Producing)	Gear	Available(Gear) — FALSE
M4	Active	Yes	ComponentC	Available(ComponentC) — TRUE

Conclusion: ComponentA, ComponentB, and ComponentC are currently available for downstream use/shipment. Gear is NOT available.

3.Is Gear Production Affected by Maintenance?

Yes. By direct substitution, Maintenance(M3) is a stated fact, and Rule 3 (Maintenance(x) → ¬Producing(x)) gives ¬Producing(M3) by Modus Ponens. Since Produces(M3, Gear) holds but Active(M3) does not hold (M3 is in Maintenance, not Active), the antecedent of Rule 2 (Produces(x,y) ∧ Active(x)) is false for x = M3, so Available(Gear) cannot be derived. Formally:

- Maintenance(M3) ... given fact
- Maintenance(M3) → ¬Producing(M3) ... Rule 3
- ∴ ¬Producing(M3) ... Modus Ponens
- ¬Active(M3) (Active and Maintenance are mutually exclusive states)
- Produces(M3, Gear) ∧ Active(M3) is FALSE (since Active(M3) is false)
- ∴ Available(Gear) is NOT entailed by the knowledge base

Therefore, Gear production is directly halted by M3's maintenance status, and this creates a supply gap for any process that depends on Gear as an input — a concrete, traceable causal chain from a single maintenance event to a downstream inventory shortfall.

4.Effectiveness of Predicate Logic in Industrial Decision-Support Systems

- Strengths — Determinism & Traceability: FOPC provides an unambiguous, auditable chain of inference (as shown above), which is valuable in safety-critical or compliance-sensitive manufacturing environments where every automated decision must be explainable.
- Strengths — Compositionality: New machines, products, and rules can be added incrementally without rewriting the whole knowledge base, supporting scalable plant-wide monitoring.
- Strengths — Real-Time Alerting: Because facts (Active/Maintenance) update from sensor/PLC feeds, the same rule set can automatically flag downstream Available(y) changes the instant a machine's status changes — enabling proactive supply-chain alerts (e.g., "Gear unavailable due to M3 maintenance").
- Limitations — Closed-World / Brittleness: Pure FOPC assumes complete, certain knowledge; it cannot natively represent partial availability, probabilistic downtime estimates, or uncertain sensor readings (e.g., a machine that is 80% likely to fail soon).
- Limitations — Scalability of Rule Sets: As the number of machines, products, and interdependencies (multi-machine assembly lines, shared components) grows, purely hand-crafted rule sets become difficult to maintain and can suffer combinatorial explosion during inference.
- Limitations — No Temporal/Quantitative Reasoning: Standard FOPC does not natively express time ("maintenance will end in 2 hours") or quantity ("only 30% of Gear stock remains"), both of which are critical for real production planning.
- Recommended Extension: Combine FOPC with probabilistic or fuzzy extensions (e.g., Markov Logic Networks) and temporal logic to retain the interpretability of predicate rules while adding the uncertainty- and time-awareness that real Industry 4.0 environments require.

Q3. Word Sense Disambiguation in E-Commerce Search Engines

1. Correct Word Sense for Each Query (Context-Based)

User Query	Sense 1	Sense 2	Clicked Result	Correct Sense
Apple accessories	Fruit	Technology Brand	iPhone Charger	Technology Brand
Mouse wireless	Animal	Computer Device	Bluetooth Mouse	Computer Device
Java tutorial	Island	Programming Language	Coding Lessons	Programming Language
Python course	Snake	Programming Language	Software Development Training	Programming Language

In every case, the sense that matches the user's subsequent click behaviour is the technology/computing sense, not the literal/biological sense — a pattern typical of e-commerce and tech-education platforms where the dominant domain skews word senses away from their most common everyday meaning.

2. Semantic Cues Used for Disambiguation

- Collocation Cues: Co-occurring words strongly bias sense selection — "accessories" next to "Apple" signals gadgets, not fruit; "wireless" next to "Mouse" signals a device, not an animal; "tutorial"/"course" next to "Java"/"Python" signal learning content, not an island or a reptile.
- Domain/Platform Context: The fact that the query occurs on an e-commerce or e-learning platform (rather than, say, a cooking or wildlife website) is itself a strong prior that shifts probability mass toward the technology sense.
- Behavioural/Click-Through Evidence: The clicked result acts as ground-truth feedback — a click on "iPhone Charger", "Bluetooth Mouse", "Coding Lessons", or "Software Development Training" retroactively confirms which sense the user intended, and this signal can be fed back into the WSD model.
- Selectional Preferences: Verbs/nouns like "tutorial", "course", and "wireless" have selectional restrictions that are only satisfiable by the technology sense (an island cannot have a "tutorial"; a snake cannot have a "course" in the educational sense), so syntactic co-occurrence itself rules out the alternate sense.
- User/Session History: Prior searches or purchase history in a session (e.g., previous queries about laptops or programming) provide additional distributional context that reinforces the technology-sense interpretation.

3. Impact of Incorrect Sense Selection on Search Engine Performance

- Irrelevant Results & Poor Recall/Precision: If "Apple" is disambiguated as Fruit, the search engine would surface grocery items instead of electronics, producing results with zero relevance to user intent.
- Increased Bounce Rate: Users faced with wrong-sense results are likely to abandon the search session quickly, which platforms interpret as a negative satisfaction signal.
- Lost Revenue / Conversion Drop: In e-commerce, sense errors directly translate into missed sales — a user searching for a computer mouse who is shown pet supplies will not convert.
- Degraded Personalization & Ranking Models: Downstream recommendation and ranking systems that rely on the (mis-)tagged sense will propagate the error, polluting click-through-rate training data and degrading future ranking quality.
- Erosion of User Trust: Repeated irrelevant results due to WSD failures reduce user confidence in the search engine, pushing users toward competitor platforms.

4.Industrial-Scale WSD Strategy for Improving Recommendation Accuracy

- Embedding-Based Contextual Disambiguation: Use contextual word embeddings (e.g., transformer-based encoders) that generate different vector representations for "Apple" depending on its surrounding words, rather than relying on a single static embedding per word.
- Knowledge-Base Grounding: Link ambiguous query terms to a structured product/knowledge ontology (e.g., a product taxonomy distinguishing "Apple Inc." from "apple (fruit)") so the platform can resolve sense via entity linking rather than surface-form matching alone.
- Click-Through Feedback Loop (Implicit Supervision): Continuously mine query–click pairs as weak-supervision labels to retrain the WSD/intent model, since click data (as seen in the table above) is a strong, low-cost, scalable signal for the correct sense.
- Domain-Adaptive Priors: Maintain platform-specific sense-frequency priors (e.g., on a tech marketplace, "Java"→Programming Language should have a much higher prior probability than "Java"→Island) and combine them with context scores.
- Session and User-Profile Context: Incorporate session history and user purchase/browse profile as additional disambiguation features, since a user who recently viewed laptops is far more likely to mean the programming-language sense of "Python".
- A/B Testing and Continuous Evaluation: Deploy WSD model updates through controlled A/B tests measuring click-through rate, conversion rate, and query abandonment rate to validate that disambiguation improvements translate into real business impact.
- Human-in-the-Loop Auditing for High-Ambiguity Terms: For terms with historically high error rates (brand names that are also common nouns), route a sample of low-confidence predictions to human review to continuously refine the model.

Q4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

1. How Syntactic Structures Contribute to Semantic Role Assignment

All four parsed sentences share a Subject–Verb–Object (SVO) syntactic template. Syntax-driven semantic role labelling maps positions in this syntactic frame onto semantic roles using the verb's argument structure (its predicate-argument frame):

Sentence	Subject (Syntax)	Verb	Object(s) (Syntax)	Mapped Semantic Roles
Doctor prescribed medicine to patient.	Doctor	prescribed	medicine, (to) patient	Doctor=Agent, medicine=Instrument/Theme, patient=Recipient
Patient reported severe headache.	Patient	reported	severe headache	Patient=Agent/Experiencer, headache=Symptom (Theme)
Nurse monitored patient continuously.	Nurse	monitored	patient	Nurse=Agent, patient=Recipient/Theme
Medicine reduced blood pressure.	Medicine	reduced	blood pressure	Medicine=Instrument (surface Agent), blood pressure=Theme/Patient(role)

The mapping works because the syntactic subject position typically aligns with the thematic Agent role for action verbs (prescribed, monitored), the direct object aligns with the Theme/Instrument being acted upon (medicine), and prepositional/indirect objects ("to patient") align with the Recipient role. Syntax therefore acts as a structural scaffold that narrows down the plausible semantic role assignment before verb-specific semantic knowledge fine-tunes it.

2. Are the Assigned Semantic Roles Appropriate?

- Doctor → Agent: Appropriate. The doctor is the volitional entity performing the prescribing action.
- Medicine → Instrument: Partially appropriate but context-dependent. In "Doctor prescribed medicine to patient," medicine is more precisely the Theme (the entity being transferred/prescribed) rather than an Instrument, since the doctor does not use the medicine to perform the prescribing — the medicine is what is prescribed. Instrument would better describe something like "a prescription pad." This role assignment should be refined to Theme for this sentence.
- Patient → Recipient: Appropriate in the context of "prescribed ... to patient" and "monitored patient" — the patient receives the action's effect.
- Headache → Symptom: This is a domain-specific (medical) role label rather than a standard thematic role; in generic semantic role terms it corresponds to Theme/Stimulus of the verb "reported." It is appropriate as a healthcare-specific role, but a general-purpose SRL system would need an additional domain-role layer to produce "Symptom" instead of the generic "Theme."
- Overall: The four roles are broadly correct at a coarse level, but the reuse of a single "Medicine" entity as "Instrument" is questionable across sentences — its role should be re-evaluated per-sentence rather than assigned a single fixed role for the entity, since "Medicine reduced blood pressure" casts medicine as more of a causal Agent/Instrument, while "prescribed medicine" casts it as a Theme.

3. Potential Errors from Incorrect Parsing

- Agent–Patient Reversal: A parser that mis-attaches the subject/object (e.g., due to a parsing ambiguity or passive-voice confusion) could swap Doctor and Patient, wrongly implying the patient performed the prescribing action — a critical safety-relevant error in a clinical record.
- Misattachment of Prepositional Phrases: In "Doctor prescribed medicine to patient," a parser could mistakenly attach "to patient" to the wrong head (e.g., interpreting it as modifying "medicine" rather than the verb's Recipient argument), corrupting the Recipient role assignment.
- Instrument/Theme Confusion: As discussed above, if the parser defaults to a fixed role for "medicine" regardless of syntactic context, it can mislabel causal relationships (e.g., failing to distinguish "medicine was prescribed" from "medicine caused a reduction"), which matters for downstream clinical reasoning such as adverse-drug-event detection.
- Coordination/Ellipsis Errors: Medical sentences often contain coordinated clauses or elliptical constructions ("prescribed medicine and advised rest") that a shallow SVO parser can mis-segment, causing role assignment to merge or drop arguments.

- Negation Scope Errors: A syntactic parser that fails to correctly scope negation ("patient did not report headache") can invert the intended meaning of the Symptom role, which is clinically dangerous if propagated into downstream diagnosis-support systems.
- Named Entity Boundary Errors: Multi-word medical terms ("severe headache", "blood pressure") must be correctly chunked as single semantic units; incorrect tokenization/chunking could split "blood" and "pressure" into separate roles, corrupting the Theme.

4. **Recommendations to Improve Syntax-Driven Semantic Analysis in Healthcare**

- Domain-Specific Treebank Training: Fine-tune the syntactic parser and SRL model on annotated clinical/medical corpora (e.g., clinical notes, prescription logs) rather than relying solely on general-domain treebanks, since medical sentence structures and terminology differ from everyday text.
- Joint Syntax–Semantics Modeling: Use joint parsing-and-SRL architectures (rather than a strict pipeline) so that semantic role plausibility can help correct syntactic attachment ambiguities and vice versa, reducing cascading errors.
- Medical Named-Entity Recognition (NER) Pre-processing: Run a clinical NER pass first to correctly chunk multi-word medical entities (drug names, symptoms, vitals) before syntactic parsing, preventing entity-boundary errors from corrupting role labels.
- Verb-Specific Frame Lexicons (e.g., PropBank/FrameNet adapted for clinical verbs): Maintain explicit argument-structure frames for common clinical verbs (prescribe, administer, monitor, report, reduce) so role assignment is grounded in verb-specific semantics rather than generic SVO heuristics — this directly resolves the Instrument-vs-Theme ambiguity identified in Section 2.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
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Date: ____________________	Date: ____________________	Date: ____________________